Zhongze Li (李中泽)

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RESEARCH PROFILE

Research on reliable and scalable stability analysis for converter-dominated power systems. My current work centers on (i) multi-timescale modeling and robust stability across EMT and electromechanical/phasor domains, and (ii) compositional and distributed stability analysis based on local device and network information.

EDUCATION

Tsinghua University, Beijing, China — Aug. 2024 - Present
Ph.D. Student, Department of Electrical Engineering
Advisor: Associate Professor Feng Liu

Tsinghua University, Beijing, China — 2020 - 2024
B.Eng., Mathematics and Physics & Electrical Engineering (Weiyang College)
（数理基础科学＋电气工程及其自动化，未央书院）

AWARDS & SCHOLARSHIPS

NR Electric Scholarship （南瑞继保奖学金）	2025
Future Scholar Scholarship （未来学者奖学金）	2024
Outstanding Graduate, Tsinghua University （清华大学优良毕业生）	2024

TEACHING

Teaching Assistant, **Principles of Automatic Control** （自动控制原理）	2024 - 2026
Teaching Assistant, **Signals, Systems and Control** （信号、系统与控制）	2024 - 2026

RESEARCH INTERESTS

Multi-timescale modeling and stability: EMT-Phasor/RMS interaction, model validity, robust stability, structured singular value (mu), frequency-response based uncertainty identification.

Compositional and distributed stability: passivity/dissipativity, local-to-global stability certificates, distributed margins, heterogeneous converter dynamics, scalable robust analysis.

SELECTED PUBLICATIONS

When Can Phasor-Domain Device Models Be Trusted for Electromechanical Stability Analysis of Grid-Forming Converter-Dominated Microgrids?
Zhongze Li, Xiaoyu Peng, Xi Ru, Zhaojian Wang, Jianxin Zhang, Yingshang Liu, Feng Liu.
arXiv:2606.08082, 2026.

How Passivity of Bus Dynamics Influences Power System Stability: A Sensitivity Analysis
Zhongze Li, Xi Ru, Xiaoyu Peng, Beisi Tan, Pengfei Gao, Feng Liu.
2025 IEEE 9th Conference on Energy Internet and Energy System Integration (EI2), 2025.

Distributed Stability Analysis for Power Systems with Grid-Following and Grid-Forming Devices
Zhongze Li, Bing Zeng, Xiaoyu Peng, Maohai Wang, Xi Ru, Yuzhi Wang, Feng Liu.
Energy Conversion and Economics Forum / IET Conference Proceedings, 2025.

SELECTED PUBLICATIONS (CONTINUED)

Toward Less Conservative Distributed Stability Analysis of Power Systems via Matrix-Valued Differential Passivity Indices
Xi Ru, Cong Fu, **Zhongze Li**, Xiaoyu Peng, Feng Liu.
arXiv:2605.04821, 2026.

Compositional Grid Codes With Guarantee on Both Stability and Dynamic Performance
Xiaoyu Peng, Cong Fu, **Zhongze Li**, Xi Ru, Zhaojian Wang, Feng Liu.
IEEE Transactions on Power Systems, 2026.

Positive Damping Region: A Graphic Tool for Passivization Analysis with Passivity Index
Xiaoyu Peng, Xi Ru, **Zhongze Li**, Jianxin Zhang, Xinghua Chen, Feng Liu.
arXiv:2601.10475, 2026.

Relative-Degree Wall Restricts Passivity-Based Stability Analysis in Inverter-Dominant Grids
Xiaoyu Peng, **Zhongze Li**, Xi Ru, Xinghua Chen, Feng Liu.
arXiv preprint, 2026.

SELECTED ACADEMIC ACTIVITIES

Presenter, The 4th Energy Conversion and Economics Annual Forum (ECE 2024), Beijing, China. Talk: “Distributed Stability Analysis for Power Systems with Grid-Following and Grid-Forming Devices.”

RESEARCH METHODS & TOOLS

Robust control and frequency-domain analysis: H-infinity methods, structured singular value (mu), passivity/dissipativity, Nyquist/Bode analysis, uncertainty modeling.

Power-system modeling and simulation: converter-dominated systems, EMT/RMS multi-timescale modeling, small-signal and nonlinear stability analysis.

Computational tools: MATLAB/Simulink, Python; numerical simulation and reproducible research workflows.

LINKS

Academic homepage: https://lizzz766.github.io
Google Scholar: https://scholar.google.com/citations?user=HKIw3YUAAAAJ&hl=en
GitHub: https://github.com/lizzz766

Last updated September 2026. For the most up-to-date publication list, please refer to Google Scholar.